Separable Differential Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solving by separation of variables**

Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.

$y\,dy = x\,dx$, so $\frac{y^2}{2} = \frac{x^2}{2} + C$. Using $y(0) = 3$: $C = \frac{9}{2}$. Solution: $y^2 = x^2 + 9$, i.e. $y = \sqrt{x^2 + 9}$ (taking the positive root since $y(0) = 3 > 0$).

2 **[Foundational]** • **Target: Growth, decay, and applications**

A population grows according to $\frac{dP}{dt} = 0.05P$, with $P(0) = 2000$.

a Solve for $P(t)$. $P(t) = 2000e^{0.05t}$.

b Find $P(20)$, to the nearest whole number. $P(20) = 2000e^{1} \approx 5437$.

(a) $P(t) = 2000e^{0.05t}$.; (b) $P(20) = 2000e^{1} \approx 5437$.

3 **[Foundational]** • **Target: Growth, decay, and applications**

Newton's law of cooling gives $\frac{dT}{dt} = -k(T - 20)$ for an object cooling in a $20°$C room. If $T(0) = 100$ and $T(5) = 60$, find k and hence $T(t)$.

Solving: $T(t) = 20 + 80e^{-kt}$. Using $T(5) = 60$: $40 = 80e^{-5k}$, so $e^{-5k} = \frac{1}{2}$, giving $k = \frac{\ln 2}{5} \approx 0.1386$. So $T(t) = 20 + 80e^{-0.1386t}$.

4 **[Foundational]** • **Target: More separable equations**

Solve $\frac{dy}{dx} = y(1 - y)$ (logistic-type equation), given $y(0) = \frac{1}{2}$, using partial fractions $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.

$\int \left(\frac{1}{y} + \frac{1}{1-y}\right) dy = \int dx$: $\ln|y| - \ln|1 - y| = x + C$, i.e. $\ln\left|\frac{y}{1-y}\right| = x + C$. Using $y(0) = \frac{1}{2}$: $\frac{y}{1-y} = 1$ at $x = 0$, so $C = 0$: $\frac{y}{1-y} = e^{x}$, giving $y = \frac{e^{x}}{1 + e^{x}}$.

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5 [Foundational] • Target: Mixing and applied models

A tank has 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution drains at the same rate, so the volume stays constant. Let $y(t)$ be the amount of salt (kg) at time t. The differential equation is $\frac{dy}{dt} = 6 - \frac{3y}{100}$.

a Find the equilibrium value of y (where $\frac{dy}{dt} = 0$). $6 = \frac{3y}{100}$ gives $y = 200$ kg.

b Explain, in words, what this equilibrium represents physically. In the long run, the salt content approaches 200 kg, at which point the rate of salt entering equals the rate leaving, keeping the amount constant.

(a) $6 = \frac{3y}{100}$ gives $y = 200$ kg.; (b) In the long run, the salt content approaches 200 kg, at which point the rate of salt entering equals the rate leaving, keeping the amount constant.

6 [Foundational] • Target: Slope fields and qualitative behavior

For the differential equation $\frac{dy}{dx} = x - y$, find the slope of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$.

At $(0, 0)$: slope $= 0 - 0 = 0$. At $(1, 0)$: slope $= 1 - 0 = 1$. At $(1, 1)$: slope $= 1 - 1 = 0$.

7 [Foundational] • Target: Slope fields and qualitative behavior

Verify that $y = x - 1 + 2e^{-x}$ is a solution to $\frac{dy}{dx} = x - y$, by substituting into both sides.

$\frac{dy}{dx} = 1 - 2e^{-x}$. Right side: $x - y = x - (x - 1 + 2e^{-x}) = 1 - 2e^{-x}$. Both sides equal $1 - 2e^{-x}$, confirming the solution.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Visualizing a logistic solution curve

The graph shows the logistic solution $y = \frac{e^x}{1 + e^x}$ found earlier for $\frac{dy}{dx} = y(1 - y)$, $y(0) = \frac{1}{2}$. Use the graph to describe the long-run behavior of y as $x \to \infty$ and as $x \to -\infty$.

As $x \to \infty$, $y \to 1$ (the curve levels off approaching the horizontal asymptote $y = 1$). As $x \to -\infty$, $y \to 0$. This matches the equilibrium solutions of $y(1 - y) = 0$ at $y = 0$ and $y = 1$.

9 [Intermediate] • Target: Another applied separable equation

Radioactive decay follows $\frac{dA}{dt} = -kA$. If a sample decays from 80 g to 20 g in 30 days, find k and the half-life of the substance.

$A(t) = 80e^{-kt}$. Using $A(30) = 20$: $\frac{20}{80} = \frac{1}{4} = e^{-30k}$, so $k = \frac{\ln 4}{30} \approx 0.0462$/day. Half-life: $e^{-kt_{1/2}} = \frac{1}{2}$ gives $t_{1/2} = \frac{\ln 2}{k} = \frac{30 \ln 2}{\ln 4} = 15$ days.

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10 **[Intermediate]** • **Target: Verifying a solution to an applied model**

For the mixing problem $\frac{dy}{dt} = 6 - \frac{3y}{100}$ with $y(0) = 0$, verify that $y(t) = 200(1 - e^{-0.03t})$ is a solution, and confirm it approaches the equilibrium 200 found earlier as $t \to \infty$.

$\frac{dy}{dt} = 200(0.03)e^{-0.03t} = 6e^{-0.03t}$. Right side: $6 - \frac{3}{100} \cdot 200(1 - e^{-0.03t}) = 6 - 6(1 - e^{-0.03t}) = 6e^{-0.03t}$. Both sides match. As $t \to \infty$, $e^{-0.03t} \to 0$, so $y(t) \to 200$, confirming the equilibrium.

11 **[Intermediate]** • **Target: Verifying a solution to an applied model**

Solve $\frac{dy}{dx} = e^{x-y}$, given $y(0) = 0$, using the fact that $e^{x-y} = e^x \cdot e^{-y}$.

$e^y\,dy = e^x\,dx$: $e^y = e^x + C$. Using $y(0) = 0$: $1 = 1 + C$, so $C = 0$: $e^y = e^x$, giving $y = x$.

12 **[Intermediate]** • **Target: Verifying a solution to an applied model**

A pond's fish population P grows according to $\frac{dP}{dt} = 0.08P\left(1 - \frac{P}{500}\right)$, a logistic model with carrying capacity 500. Without solving the equation explicitly, determine the equilibrium population levels (where $\frac{dP}{dt} = 0$), and state which one is realistic for a growing population starting from $P(0) = 50$.

Setting $\frac{dP}{dt} = 0$: $P = 0$ or $P = 500$. Starting from $P(0) = 50 > 0$, the population will grow toward the stable equilibrium $P = 500$ (the carrying capacity), since $P = 0$ is an unstable equilibrium for any positive starting population.

13 **[Intermediate]** • **Target: Verifying a solution to an applied model**

Solve $\frac{dy}{dx} = \frac{\cos x}{y}$, given $y(0) = -2$, and state why the negative root must be chosen.

$y\,dy = \cos x\,dx$: $\frac{y^2}{2} = \sin x + C$. Using $y(0) = -2$: $2 = 0 + C$, so $C = 2$: $y^2 = 2\sin x + 4$, giving $y = -\sqrt{2\sin x + 4}$. The negative root is required because $y(0) = -2 < 0$, and y must vary continuously from that starting sign.

14 **[Intermediate]** • **Target: A final applied separable model**

A drug is eliminated from the bloodstream according to $\frac{dC}{dt} = -0.2C$, where C is concentration (mg/L) and t is hours. If the initial dose gives $C(0) = 50$ mg/L, find $C(t)$, then find how many hours it takes for the concentration to drop below 5 mg/L, to one decimal place.

$C(t) = 50e^{-0.2t}$. Setting $50e^{-0.2t} = 5$: $e^{-0.2t} = 0.1$, so $t = \frac{\ln 0.1}{-0.2} = \frac{-2.3026}{-0.2} \approx 11.5$ hours.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Slope fields and qualitative behavior • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Level 2 · MediumFind the particular solution to the differential equation $\frac{dy}{dx} = \frac{x}{y}$ with initial condition $y(0) = -4$. State the domain of the solution.

1. Separate variables: $y\,dy = x\,dx$ 2. Integrate both sides: $\int y\,dy = \int x\,dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C_1 \Rightarrow y^2 = x^2 + C$ 3. Substitute initial condition $x = 0, y = -4$: $(-4)^2 = 0^2 + C \Rightarrow C = 16 \Rightarrow y^2 = x^2 + 16$ 4. Solve for y: Since $y(0) = -40$ for all x, y is strictly negative and never zero. The domain is $(-\infty, \infty)$.

16 [Challenging] • Target: Visualizing a logistic solution curve • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Level 1 · FoundationsWhich of the following functions is a solution to the differential equation $y'' + 4y = 0$? (A) $y = e^{2x}$ (B) $y = \sin(2x) - 3\cos(2x)$ (C) $y = 4\cos(4x)$ (D) $y = x^2 - 4$

Test (B): $y = \sin(2x) - 3\cos(2x)$. First derivative: $y' = 2\cos(2x) + 6\sin(2x)$ Second derivative: $y'' = -4\sin(2x) + 12\cos(2x) = -4(\sin(2x) - 3\cos(2x)) = -4y$ Substitute into the differential equation: $y'' + 4y = -4y + 4y = 0$ The equation is satisfied identically. Correct answer is (B).

17 [Challenging] • Target: Another applied separable equation • Cross-Lesson: 6.01 Related Rates

Level 3 · Advanced / AP FRQLet $y = f(x)$ be the particular solution to $\frac{dy}{dx} = xy^2$ with $f(1) = -1$. Use Euler's method with step size $\Delta x = 0.5$ starting at $x = 1$ to approximate $f(2)$, and determine whether the approximation is an overestimate or underestimate by computing $\frac{d^2y}{dx^2}$ at $(1, -1)$.

Step 1 · Euler Iteration: • $(x_0, y_0) = (1, -1)$. Slope $m_0 = (1)(-1)^2 = 1$. $y_1 = -1 + (1)(0.5) = -0.5$. Point $(1.5, -0.5)$. • $(x_1, y_1) = (1.5, -0.5)$. Slope $m_1 = (1.5)(-0.5)^2 = 1.5(0.25) = 0.375$. $y_2 = -0.5 + (0.375)(0.5) = -0.5 + 0.1875 =$ **−0.3125**. Step 2 · Second Derivative Evaluation: $\frac{d^2y}{dx^2} = \frac{d}{dx}[xy^2] = (1)y^2 + x(2yy') = y^2 + 2xy(xy^2) = y^2 + 2x^2y^3$. At $(1, -1)$: \[\left. \frac{d^2y}{dx^2} \right|_{(1, -1)} = (-1)^2 + 2(1)^2(-1)^3 = 1 + 2(-1) = 1 - 2 =.

18 [Challenging] • Target: Separable equations with algebraic solutions • Cross-Lesson: 6.02 Optimization and Curve Sketching

Solve the separable DE $\frac{dy}{dx} = \frac{x+1}{y}$ given $y = 2$ when $x = 0$.

Separate: $y\,dy = (x+1)\,dx$. Integrate: $\frac{y^2}{2} = \frac{x^2}{2} + x + c$. At $(0, 2)$: $2 = c$. So $y^2 = x^2 + 2x + 4$.

19 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 7.01 Integration by Substitution

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

Separate variables and integrate: $\int \frac{1}{\theta - 20}\,d\theta = -\int k\,dt \Rightarrow \ln(\theta - 20) = -kt + C$ Apply initial condition $t = 0, \theta = 80$: $\ln(60) = C \Rightarrow \ln\left(\frac{\theta - 20}{60}\right) = -kt \Rightarrow \theta - 20 = 60e^{-kt}$ Apply condition $t = 10, \theta = 50$: $30 = 60e^{-10k} \Rightarrow e^{-10k} = \frac{1}{2} \Rightarrow k = \frac{\ln 2}{10} \approx 0.06931$ Set $\theta = 30$.

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20 [Challenging] • Target: A final applied separable model • Cross-Lesson: 7.02 Integration by Parts

Solve the equation $z^3 + 8 = 0$, expressing all solutions in Cartesian form $a + bi$.

Write $z^3 = -8 = 8e^{i\pi}$. By De Moivre's Theorem, the three cube roots are: $z_k = 2\exp\left(i\frac{\pi + 2\pi k}{3}\right)$ for $k = 0, 1, 2$. For $k = 0$: $z_0 = 2e^{i\pi/3} = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 1 + i\sqrt{3}$. For $k = 1$: $z_1 = 2e^{i\pi} = 2(-1) = -2$. For $k = 2$: $z_2 = 2e^{i5\pi/3} = 2e^{-i\pi/3} = 2\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = 1 - i\sqrt{3}$. Notice that $1 \pm i\sqrt{3}$ are complex conjugates, as predicted by the Conjugate Root Theor.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving by separation of variables

A population grows according to the logistic model $\frac{dP}{dt} = 0.08P\left(1 - \frac{P}{315}\right)$. Find the population P at the point of maximum growth.

a. 630

b. 78.75

c. 315

d. **157.5** ✓

Logistic growth is maximal at the point of inflection, where $P = M/2 = 315/2 = 157.5$.

22 [Foundational] • Target: Growth, decay, and applications

Consider the differential equation $\frac{dy}{dx} = 2x$. Which of the following is a particular solution passing through the point $(1, 3)$?

a. $y = x^2 + 2$ ✓

b. $y = x^2 + 1$

c. $y = 2x + 1$

d. $y = 2x^2 - 1$

The given differential equation is $\frac{dy}{dx} = 2x$. To find the general solution, we integrate both sides with respect to x: $\int \frac{dy}{dx}dx = \int 2x dx$ $y = x^2 + C$, where C is the constant of integration. We are given that the particular solution passes through the point $(1, 3)$. Substitute these values into the general solution: $3 = (1)^2 + C$ $3 = 1 + C$ $C = 3 - 1 = 2$. Therefore, the particular solution is $y = x^2 + 2$.

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23 [Foundational] • Target: More separable equations

Exponential decay: $\frac{dN}{dt} = -kN$, $k > 0$. Solution:

a. $N = N_0 e^{kt}$

b. $N = N_0/(1 + kt)$

c. $N = N_0 - kt$

d. $N = N_0 e^{-kt}$ ✓

Applying Differential Equations principles, $N = N_0 e^{-kt}$. Therefore, the verified correct option is $N = N_0 e^{-kt}$.

24 [Foundational] • Target: Mixing and applied models

Separable ODE $\frac{dy}{dx} = xy$. Solution:

a. $y = x^2/2 + C$

b. $y = Ce^x$

c. $y = Ce^{x^2/2}$ ✓

d. $y = Ce^{x^2}$

$\frac{dy}{y} = x\,dx \Rightarrow \ln|y| = x^2/2 + C$.

25 [Foundational] • Target: Slope fields and qualitative behavior

Separable ODE: $\frac{dy}{dx} = \frac{x}{y}$. Solution (implicit):

a. $x^2 + y^2 = C$

b. $y^2 = x^2$

c. $y = x + C$

d. $y^2 = x^2 + C$ ✓

$y\,dy = x\,dx \Rightarrow y^2/2 = x^2/2 + C_1$.

26 [Foundational] • Target: Visualizing a logistic solution curve

Logistic equation steady states (equilibria):

a. $P = 0$ only

b. $P = K$ only

c. $P = 0$ **and** $P = K$ ✓

d. $P = K/2$

$rP(1 - P/K) = 0 \Rightarrow P = 0$ or $P = K$.

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27 [Foundational] • Target: Another applied separable equation

The slope field for a differential equation is shown above. The differential equation is of the form $\frac{dy}{dx} = ky$ for some constant k. If the solution curve passes through the point $(0, 1)$ and $(1, 2)$, find the value of k.

a. $\ln 2$ ✓

b. 2

c. $\frac{1}{2}$

d. 1

The differential equation is given by $\frac{dy}{dx} = ky$. This is a separable differential equation. We can rewrite it as $\frac{dy}{y} = k\,dx$. Integrating both sides, we get $\int \frac{dy}{y} = \int k\,dx$, which yields $\ln|y| = kx + C_1$. Exponentiating both sides, we have $|y| = e^{kx + C_1} = e^{C_1}e^{kx}$. Let $C = \pm e^{C_1}$, so $y = Ce^{kx}$. We are given that the solution curve passes through the point $(0, 1)$. Substituting these values into the general solution: $1 = Ce^{k(0)}$ $1 = Ce^0$.

28 [Foundational] • Target: Separable equations with algebraic solutions

Consider the slope field shown below. Which of the following differential equations could represent this slope field, assuming the slopes are defined at all points except where division by zero occurs? $\frac{dy}{dx} = \frac{x}{y}$

a. $\frac{dy}{dx} = \frac{x}{y}$ ✓

b. $\frac{dy}{dx} = \frac{y}{x}$

c. $\frac{dy}{dx} = x - y$

d. $\frac{dy}{dx} = x + y$

The slope field shows that at points where $y = 0$ (the x-axis), the slopes are undefined (vertical lines). This is consistent with the differential equation $\frac{dy}{dx} = \frac{x}{y}$, as division by zero occurs when $y = 0$. Additionally, observe the slopes in different quadrants. In the first quadrant ($x > 0, y > 0$), the slopes are positive, consistent with $\frac{x}{y} > 0$. In the second quadrant ($x0$), the slopes are negative, consistent with $\frac{x}{y}0$. In the fourth quadrant ($x > 0, y < 0$), the slopes are.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Verifying a solution to an applied model

A population of bacteria follows a logistic growth model where $P(t)$ is the population at time t (in hours). The differential equation governing the population growth is $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{1000}\right)$. If the initial population is $P(0) = 200$, use Euler's method with a step size of $\Delta t = 1$ hour to approximate the population after 1 hour, $P(1)$.

a. 210

b. **208** ✓

c. 212

d. 209

Euler's method approximates the next value $P(t + \Delta t)$ using the formula $P(t + \Delta t) \approx P(t) + \frac{dP}{dt}|_t \cdot \Delta t$. We are given $P(0) = 200$ and $\Delta t = 1$. First, we find the rate of change at $t = 0$: $\frac{dP}{dt}|_{t=0} = 0.05 \cdot P(0)\left(1 - \frac{P(0)}{1000}\right)$ $\frac{dP}{dt}|_{t=0} = 0.05 \cdot 200\left(1 - \frac{200}{1000}\right)$ $\frac{dP}{dt}|_{t=0} = 10(1 - 0.2)$ $\frac{dP}{dt}|_{t=0} = 10 \cdot 0.8$ $\frac{dP}{dt}|_{t=0} = 8$ Now, we.

30 [Intermediate] • Target: A final applied separable model

Identify the general solution to the first-order differential equation $\frac{dy}{dx} = 3y$.

a. $y = C\ln(3x)$

b. $y = \frac{3}{2}x^2 + C$

c. $y = Ce^{-3x}$

d. $y = Ce^{3x}$ ✓

Separation of variables $\frac{dy}{y} = 3dx \Rightarrow \ln|y| = 3x + c_1 \Rightarrow y = Ce^{3x}$.

31 [Intermediate] • Target: Solving by separation of variables

For the differential equation $\frac{dy}{dx} = 1x - 5y$, find the slope of the solution curve at the point $(3, 2)$.

a. -9

b. -5

c. 7

d. **-7** ✓

Substitute $x = 3$, $y = 2$: $\frac{dy}{dx} = 1(3) - 5(2) = -7$.

32 [Intermediate] • Target: Growth, decay, and applications

A second-order ODE with repeated root $r = r_1$ has general solution:

a. $y = C_1e^{r_1x} + C_2e^{-r_1x}$

b. $y = C_1e^{r_1x}$

c. $y = (C_1 + C_2)e^{2r_1x}$

d. $y = (C_1 + C_2x)e^{r_1x}$ ✓

Applying Differential Equations principles, Repeated root: multiply by x. Therefore, the verified correct option is $y = (C_1 + C_2x)e^{r_1x}$.

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33 [Intermediate] • Target: More separable equations

For the autonomous equation $y' = -y^2 - y$, which equilibrium is unstable?

a. $y = -1$ ✓

b. $y = 0$

c. both are stable

d. neither is stable

Equilibria: $y = -1$, $y = 0$. Checking the sign of y' on each side (phase line), solutions move toward $y = 0$ and away from $y = -1$.

34 [Intermediate] • Target: Mixing and applied models

For the autonomous equation $y' = -y^2 + 5y$, which equilibrium is stable?

a. $y = 5$ ✓

b. $y = 0$

c. both are stable

d. neither is stable

Equilibria: $y = 0$, $y = 5$. Checking the sign of y' on each side (phase line), solutions move toward $y = 5$ and away from $y = 0$.

35 [Intermediate] • Target: Slope fields and qualitative behavior

Bernoulli equation $y' + P(x)y = Q(x)y^n$: substitution to linearise is:

a. $v = y^n$

b. $v = y^{1-n}$ ✓

c. $v = y^2$

d. $v = 1/y$

Applying Differential Equations principles, $v = y^{1-n}$ linearises Bernoulli. Therefore, the verified correct option is $v = y^{1-n}$.

36 [Intermediate] • Target: Visualizing a logistic solution curve

Homogeneous 2nd-order: characteristic equation $r^2 + r - 6 = 0$. Roots:

a. $r = 2, -3$ ✓

b. $r = -2, 3$

c. $r = 1, -6$

d. $r = 3, 2$

Applying Differential Equations principles, $(r + 3)(r - 2) = 0 \Rightarrow r = 2, -3$. Therefore, the verified correct option is $r = 2, -3$.

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37 [Intermediate] • Target: Another applied separable equation

Second-order constant-coefficient ODE $y'' - y = 0$: characteristic equation:

a. $r - 1 = 0$

b. $r^2 + 1 = 0$

c. $r^2 - 1 = 0$ ✓

d. $r^2 + r - 1 = 0$

Applying Differential Equations principles, $r^2 - 1 = 0 \Rightarrow r = \pm 1$. Therefore, the verified correct option is $r^2 - 1 = 0$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Separable equations with algebraic solutions • Cross-Lesson: 6.02 Optimization and Curve Sketching

Radioactive decay: A has a 10-year half-life. After 30 years, fraction remaining:

a. 1/4

b. 1/8 ✓

c. 1/6

d. 1/3

Applying Differential Equations principles, Three half-lives: $(1/2)^3 = 1/8$. Therefore, the verified correct option is $1/8$.

39 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 7.01 Integration by Substitution

Linear homogeneous 2nd-order ODE: if y_1, y_2 are solutions, so is:

a. y_1^2

b. $y_1 y_2$

c. y_1'

d. $y_1 + y_2$ ✓

Applying Differential Equations principles, Superposition principle. Therefore, the verified correct option is $y_1 + y_2$.

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40 [Challenging] • Target: A final applied separable model • Cross-Lesson: 7.02 Integration by Parts

First-order linear ODE solution structure: general solution = homogeneous solution + ____

a. particular solution ✓

b. another homogeneous solution

c. the initial condition

d. zero

Applying Differential Equations principles, $y = y_h + y_p$. Therefore, the verified correct option is particular solution.

41 [Challenging] • Target: Solving by separation of variables • Cross-Lesson: 8.01 Area Between Curves

General solution of $\frac{dy}{dx} = 2x$:

a. $y = x^2$

b. $y = 2$

c. $y = x^2 + C$ ✓

d. $y = 2x + C$

Applying Differential Equations principles, Integrate: $y = x^2 + C$. Therefore, the verified correct option is $y = x^2 + C$.

42 [Challenging] • Target: Growth, decay, and applications • Cross-Lesson: 8.02 Volumes of Revolution

Initial value problem: $y' = y$, $y(0) = 2$. Solution:

a. $y = e^{2x}$

b. $y = 2e^x$ ✓

c. $y = 2x + 2$

d. $y = xe^x$

Applying Differential Equations principles, $y = Ce^x$; $y(0) = C = 2$. Therefore, the verified correct option is $y = 2e^x$.

43 [Challenging] • Target: More separable equations • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Method of undetermined coefficients: for $y'' + y = e^x$, try particular solution:

a. $y_p = Ae^x$ ✓

b. $y_p = Ae^{2x}$

c. $y_p = Axe^x$

d. $y_p = A\sin x$

e^x is not a solution of the homogeneous equation; try Ae^x.

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44 [Challenging] • Target: Mixing and applied models • Cross-Lesson: 1.01 Precalculus Review for Calculus

Resonance in $x'' + \omega_0^2 x = F_0 \cos(\omega_0 t)$: particular solution grows as:

a. $\sin(\omega_0 t)$

b. $e^{\omega_0 t}$

c. $t^2 \cos(\omega_0 t)$

d. $t\cos(\omega_0 t)$ ✓

Resonance: particular solution $\propto t\sin$ or $t\cos$.

45 [Challenging] • Target: Slope fields and qualitative behavior • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Cauchy-Euler $x^2 y'' + xy' + y = 0$: characteristic roots $r = \pm i$. Solution:

a. $y = C_1 x^i + C_2 x^{-i}$

b. $y = C_1 \cos(\ln x) + C_2 \sin(\ln x)$ ✓

c. $y = C_1 \ln x + C_2$

d. $y = x(C_1 \cos x + C_2 \sin x)$

Complex roots: $y = C_1 \cos(\ln x) + C_2 \sin(\ln x)$.

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